

Genghan Zhang

zhang677.github.io | ghzhang19@gmail.com

EDUCATION

Stanford University

PhD Student in Computer Science

September 2023 - May 2028 (expected)

Stanford, USA

- Research Interests: Self-improving LLM Agents for Programming AI Accelerators

Tsinghua University

Bachelor of Engineer in Electronic Information Science and Technology

August 2019 - June 2023

Beijing, China

- GPA: 3.94/4.00 (Top 3%)

RESEARCH EXPERIENCE

Research Assistant

Department of Computer Science, Stanford University

April 2024 - Present

Stanford, CA

- Advisor: Prof. Kunle Olukotun
- Designing self-improving LLM agents for AI accelerator kernel implementation and optimization.

Research Assistant

Department of Computer Science, Stanford University

January 2024 - March 2024

Stanford, CA

- Advisor: Prof. Azalia Mirhoseini
- Proposed GPU kernel fusion techniques to accelerate FFN layers for LLM inference by utilizing the sparsity of activation. *CATS: Context-Aware Thresholding for Sparsity in Large Language Models* was accepted by COLM 2024

Research Assistant

Department of Computer Science, Stanford University

March 2022 - December 2023

Remote

- Advisor: Prof. Fredrik Kjølstad
- Designed an algorithm template and code generation algorithm for *sparse workspace* to solve the sparse scattering problem with a sparse tensor algebra compiler called TACO. Accepted by PLDI 2024.

SELECTED PUBLICATIONS

- **AccelOpt: A Self-Improving LLM Agentic System for AI Accelerator Kernel Optimization**
Genghan Zhang, Shaowei Zhu, Anjiang Wei, Zhenyu Song, Allen Nie, Zhen Jia, Nandita Vijaykumar, Yida Wang, Kunle Olukotun.
Conference on Machine Learning and Systems (MLSys), 2026
- **Streaming Tensor Programs: A Streaming Abstraction for Dynamic Parallelism**
Gina Sohn, Genghan Zhang, Konstantin Hossfeld, Jungwoo Kim, Nathan Sobotka, Nathan Zhang, Olivia Hsu, Kunle Olukotun.
ACM International Conference on Architectural Support for Programming Languages and Operating Systems (ASPLOS), 2026
- **Adaptive Self-improvement LLM Agentic System for ML Library Development**
Genghan Zhang, Weixin Liang, Olivia Hsu, Kunle Olukotun.
International Conference on Machine Learning (ICML), 2025
- **Compilation of Modular and General Sparse Workspaces**
Genghan Zhang, Olivia Hsu, Fredrik Kjolstad.
Programming Language Design and Implementation (PLDI), 2024
- **Sgap: Towards Efficient Sparse Tensor Algebra Compilation for GPU**
Genghan Zhang, Yuetong Zhao, Yanting Tao, Zhongming Yu, Guohao Dai, Sitao Huang, Yuan Wen, Pavlos Petoumenos, Yu Wang.
CCF Transactions on High Performance Computing, 2023

WORK EXPERIENCE

Applied Scientist Intern

Amazon

June 2025 - September 2025

Santa Clara, USA

- Mentors: Shaowei Zhu and Zhenyu Song
- Self-improving LLM agentic system for AWS Trainium kernel optimization

Software Engineer

NVIDIA

June 2024 - September 2024

Santa Clara, USA

- Mentor: Andrew Kerr
- Compiler for Tile IR

Software Engineer

Infinigence Tech

May 2023 - July 2023

Beijing, China

- Mentor: Prof. Xiuhong Li (PKU)
- Assembled an in-house GPU kernel library for LLM inference, which demos the company's first-generation product.

SERVICE

- Reviewer: ICML 2025, ICLR 2025 (Notable Reviewer), NeurIPS 2024, ICLR 2025 DL4C Workshop, NeurIPS 2024 Sys2-Reasoning Workshop, NeurIPS 2022 GLFrontiers Workshop
- Artifact Evaluation Committee: ASPLOS 2025 Summer, PLDI 2025
- Program Committee: LATTE 2025, LATTE 2026

TEACHING

- Stanford CS149 - Parallel Computing (Fall 2025)

TECHNICAL SKILLS

Programming Languages & Software Tools

- Most experienced: CUDA, Python, PyTorch, Matlab
- Some experience: C++, Rust, Verilog HDL, LtSpice